

{w ∈ {A, O}* | jedem A in w folgt direkt ein O}

1

RegExp ⇒ Automat

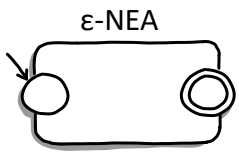
Regulärer Ausdruck =

1. einfach:
   - c ∈ Σ (Zeichen)
   - ε (Leerstring)
   - Ø (leere Menge)
2. zusammengesetzt:
   - x|y (Vereinigung)
   - xy (Konkatenation)
   - x* (kleenesche Hülle)
3. das sind alle!

vollständige Induktion

1. Beweise A(1)
2. Beweise A(n) ⇒ A(n+1)

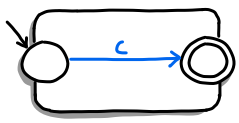
dann folgt:
A(n) für alle n ∈ $\mathbb{N}$



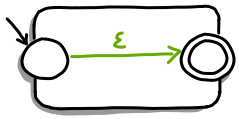
- nur ein Endzustand
- Start ≠ End

1. einfache reguläre Ausdrücke

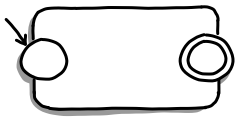
- c ∈ Σ



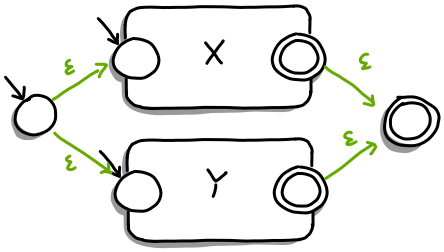
- ε



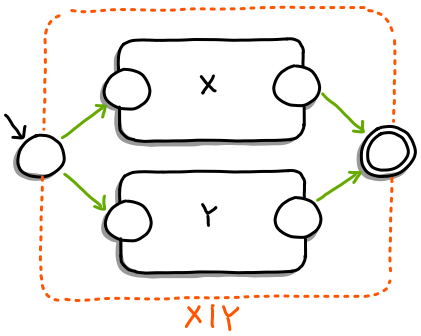
- ∅



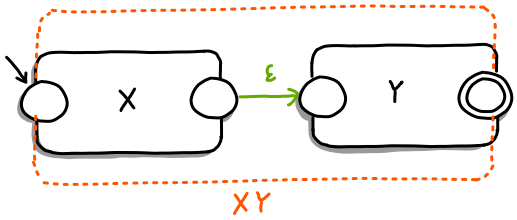
2.a Vereinigung



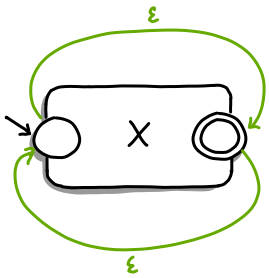
2.a Vereinigung



2.b Konkatenation

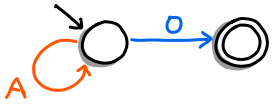


2.c Kleenesche Hülle

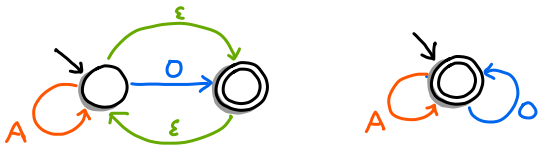




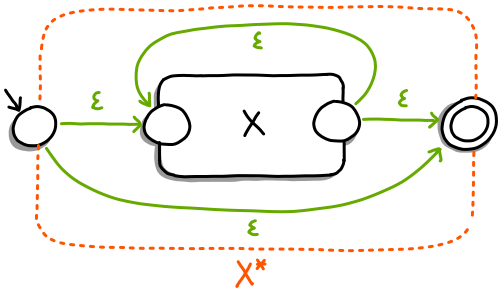
Beispiel:



{w ∈ {A, O}* | w endet auf O}

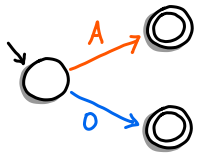


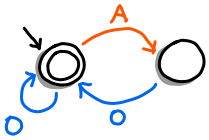
{A, O}*

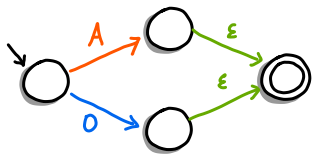


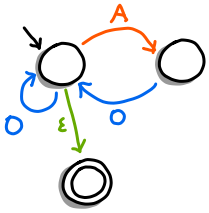
② Automat ⇒ RegExp

Schritt 1: genau ein Endzustand ≠ Start

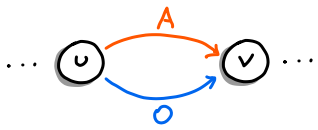








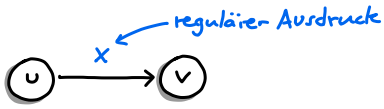
Schritt 2: Kanten zusammen fassen



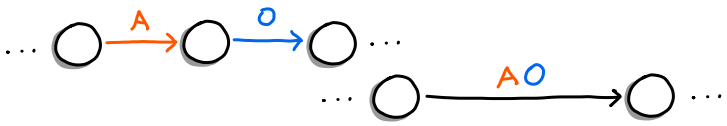


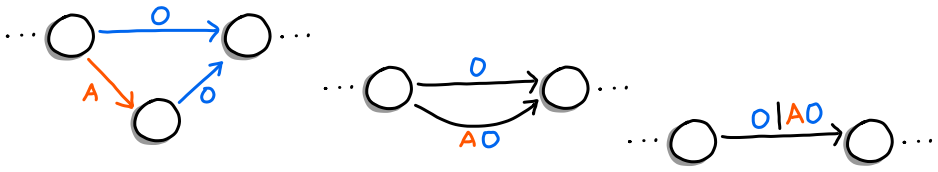
Übergänge mit regulären Ausdrücken

- nur ein Übergang pro Zustandspaar ⟨u, v⟩
- regulärer Ausdruck am Übergang ⟨u, v⟩
  = alle Strings, die man bei einem *direkten* Wechsel vom Zustand u in Zustand v einlesen darf.

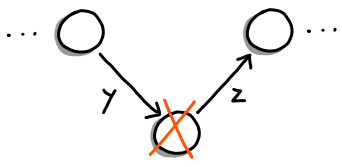


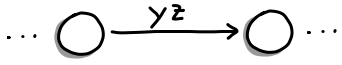
Schritt 3: innere Knoten eliminieren

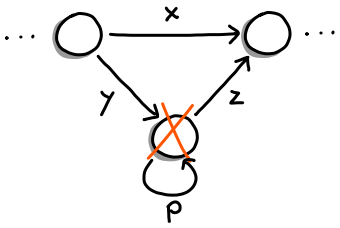


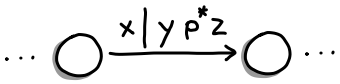


allgemein:

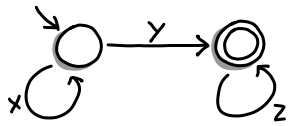








Schritt 4: Regulären Ausruck bestimmen



Ergebnis: x*yz*

Man kann endliche Automaten und reguläre Ausdrücke ineinander umwandeln.

Endliche Automaten und reguläre Ausdrücke beschreiben die gleichen Sprachen.

reguläre Sprachen

("Sprache" = Menge von Strings)